Mobility of a conserved tyrosine residue controls isoform-dependent enzyme-inhibitor interactions in nitric oxide synthases

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Table S1. Crystallographic data collection and refinement statistics

Data set [1]	eNOS-**2**	eNOS-**3**	eNOS-**4**	eNOS-**5**
Data collection				
PDB code	4CWV	4CWW	4CWX	4CWY
Space group	$P2_12_12_1$	$P2_12_12_1$	$P2_12_12_1$	$P2_12_12_1$
Cell dimensions a, b, c (Å)	58.3 106.7 156.9	58.0 106.8 156.8	57.9 106.5 157.0	57.9 106.5 157.0
Resolution (Å)	2.34 (2.38-2.34)	2.16 (2.20-2.16)	2.15 (2.19 -2.15)	2.15 (2.19-2.15)
Rmerge	0.067 (0.654)	0.064 (0.460)	0.073 (0.666)	0.071 (0.619)
I / σI	23.0 (2.1)	24.7 (2.7)	31.8 (2.3)	32.7 (2.3)
No. unique reflections	41,977	52,912	53,868	53,726
Completeness (%)	99.7 (100.0)	99.5 (99.8)	99.9 (100.0)	99.8 (100.0)
Redundancy	4.0 (4.1)	4.0 (3.9)	5.7 (4.1)	5.7 (4.1)
Refinement				
Resolution (Å)	2.34	2.16	2.15	2.15
No. reflections used	39,696	50,039	51038	50,915
R_{work} / R_{free} [2]	0.177/0.224	0.171/0.212	0.171/0.210	0.167/0.204
No. atoms				
Protein	6407	6474	6418	6407
Ligand/ion	187	197	197	202
Water	196	328	299	290
R.m.s. deviations				
Bond lengths (Å)	0.015	0.015	0.010	0.010
Bond angles (deg)	1.54	1.48	1.93	1.94

Data set [1]	eNOS-	eNOS-	eNOS-	eNOS-

	Y477A-**3**	Y477A-**6**	L111A-**3**	L111A-**5**
Data collection				
PDB code	4CWZ	4CX0	4CX1	4CX2
Space group	$P2_12_12_1$	$P2_12_12_1$	$P2_12_12_1$	$P2_12_12_1$
Cell dimensions a, b, c (Å)	57.8 106.4 156.7	57.7 106.5 156.2	57.7 106.2 156.1	57.9 106.6 156.8
Resolution (Å)	2.08 (2.12-2.08)	2.20 (2.24-2.20)	2.13 (2.17-2.13)	2.05 (2.09-2.05)
Rmerge	0.071 (0.798)	0.082 (0.614)	0.083 (0.534)	0.070 (0.556)
I / σI	20.5 (2.0)	16.9 (2.1)	16.1 (1.7)	21.0 (1.9)
No. unique reflections	59,073	50,206	54,328	59,766
Completeness (%)	99.8 (100.0)	99.4 (97.9)	99.0 (100.0)	95.7 (92.4)
Redundancy	3.7 (3.7)	3.6 (3.5)	3.6 (3.6)	3.5 (3.5)
Refinement				
Resolution (Å)	2.08	2.20	2.13	2.05
No. reflections used	55,954	47,001	51,281	56,742
R_{work} / R_{free} [2]	0.154/0.197	0.155/0.203	0.172/0.214	0.163/0.201
No. atoms				
Protein	6447	6437	6439	6435
Ligand/ion	197	207	196	199
Water	537	427	362	483
R.m.s. deviations				
Bond lengths (Å)	0.011	0.012	0.011	0.009
Bond angles (deg)	1.54	1.56	2.03	1.92

Data set [1]	nNOS-M336V/D597N-**2**	nNOS-M336V/D597N-**3**	nNOS-H341L-**2**	nNOS-H341L-**6**	Human iNOS-**7**
Data collection					
PDB code	4CX3	4CX4	4CX5	4CX6	4CX7
Space group	$P2_12_12_1$	$P2_12_12_1$	$P2_12_12_1$	$P2_12_12_1$	$P4_32_12$
Cell dimensions a, b, c (Å)	51.9 110.7 164.3	51.8 110.6 164.4	52.1 110.7 164.2	52.0 110.7 164.4	189.2 189.2 223.8
Resolution (Å)	1.97 (2.00-1.97)	1.98 (2.01-1.98)	1.80 (1.83-1.80)	1.90 (1.93-1.90)	3.16 (3.33-3.16)
Rmerge	0.041 (0.313)	0.051 (0.419)	0.055 (0.642)	0.057 (0.714)	0.232 (1.863) 0.070 (0.582) 99.6 (49.3)
I / σI	34.6 (4.7)	33.0 (3.6)	27.1 (2.1)	25.9 (2.4)	12.6 (1.5)
No. unique reflections	67,100	66,365	88,511	75,577	72,855
Completeness (%)	98.5 (91.9)	99.1 (99.9)	99.7 (100.0)	99.6 (99.9)	99.8 (98.9)
Redundancy	4.0 (3.1)	4.0 (4.0)	4.0 (4.0)	4.0 (4.0)	11.9 (10.9)
Refinement					
Resolution (Å)	1.97	1.98	1.80	1.90	3.16
No. reflections used	63,545	62,884	84,013	71,736	68,962
R_{work} / R_{free} [2]	0.175/0.209	0.175/0.212	0.191/0.226	0.185/0.217	0.173/0.216
No. atoms					
Protein	6689	6691	6685	6674	13,468
Ligand/ion	194	227	181	181	421
Water	428	413	358	382	0
R.m.s. deviations					
Bond lengths (Å)	0.014	0.015	0.011	0.010	0.014
Bond angles (deg)	1.39	1.50	1.43	1.33	1.95

[1] See Table 1 for the inhibitor chemical formula.

[2] R_{free} was calculated with the 5% of reflections set aside throughout the refinement. The set of reflections for the R_{free} calculation were kept the same for all data sets of each isoform according to those used in the data of the starting model.

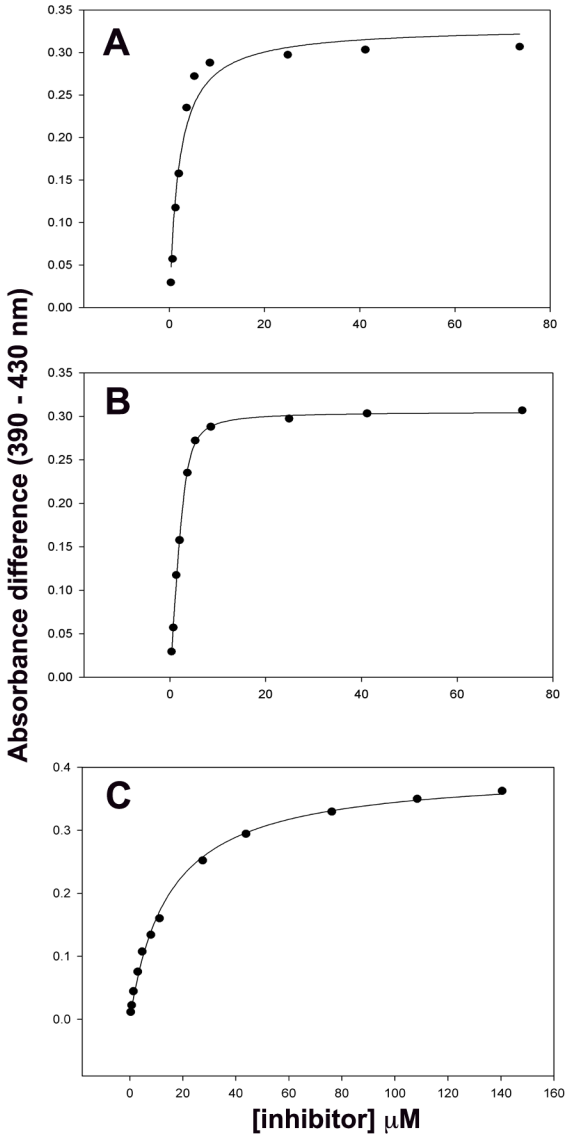


Fig. S1 Comparison of the hyperbolic and quadratic curve fittings using SigmaPlot. (A) An apparent *Ks* of 1.97 μM was given by fitting (R = 0.996) the titration data of *(2R, 4S)*-**6** to the wild type nNOS with a hyperbolic equation, $\Delta A = B_{max}[L]/(Ks + [L])$, where ΔA is the absorbance difference, B_{max} is the maximum absorbance change to infinite ligand concentration, and [L] the ligand concentration. (B) An apparent *Ks* of 0.33 μM was given by a better fitting (R = 0.999) to the same titration data as in (A) but with the

quadratic equation,[1] $\Delta A = A_0 + (B_{max}/2[E])((Ks + [E] + [L]) - ((Ks + [E] + [L])^2 - 4[E][L])^{1/2})$, where [E] is the total enzyme concentration and A_0 a constant. (C) Titration of *(3R, 4R)*-**2** to the nNOS M336V/D597N mutant can be fit vary well (R = 0.999?) with the hyperbolic equation to give an apparent *Ks* of 14.6 m M.

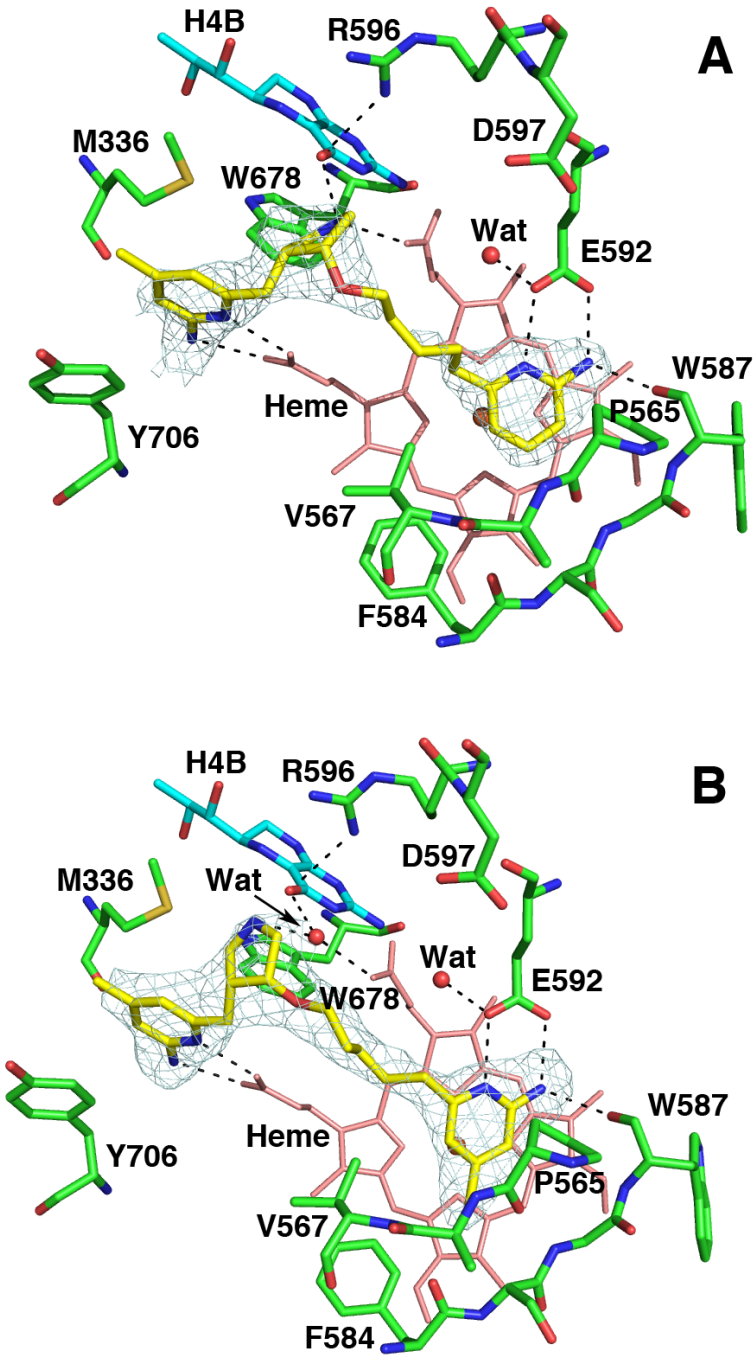


Fig. S2 The rat nNOS active site bound with (A) (3*R*, 4*R*)-**4** (3UFW, [2]) and (B) (3*R*, 4*R*)-**5** (4EUX, [2]). The omit Fo – Fc density map for each inhibitor is shown at the 2.5 σ contour level. Major hydrogen bonds are depicted with dashed lines.

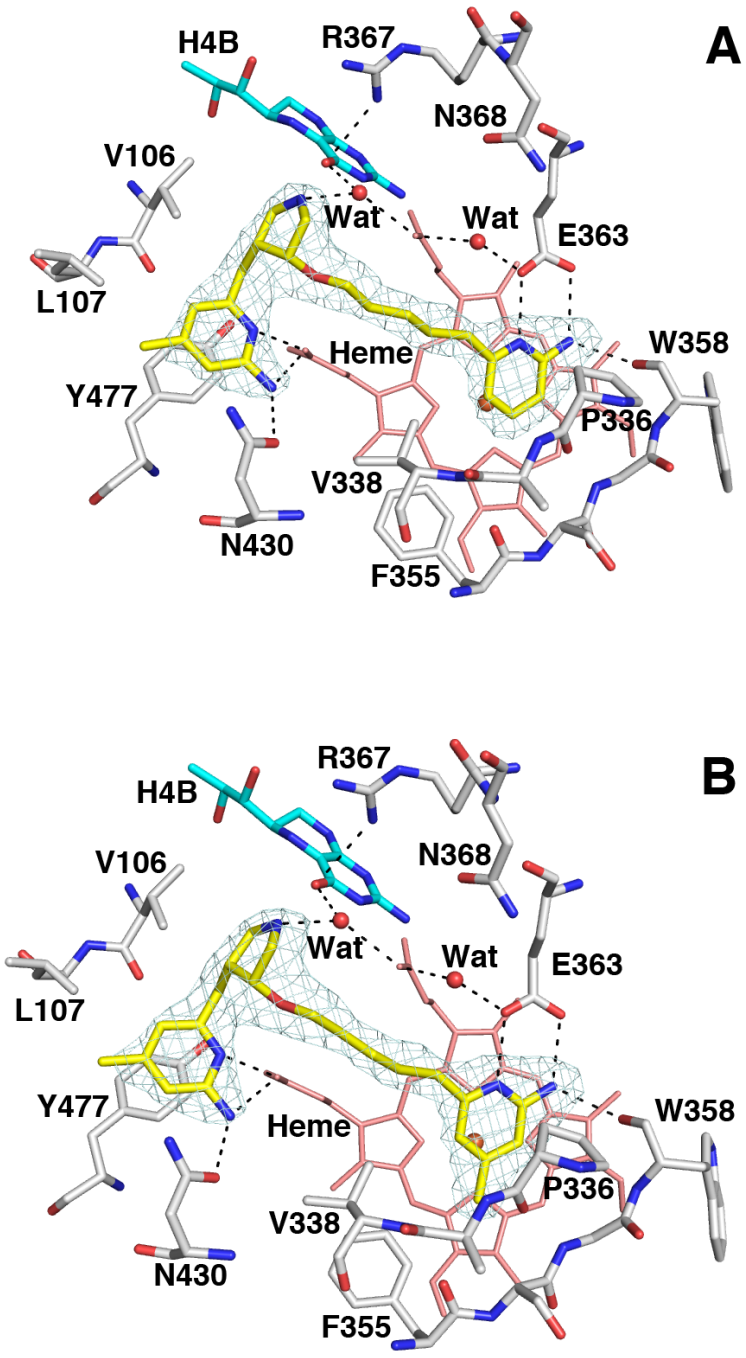


Fig. S3 The eNOS active site bound with (A) (3*R*, 4*R*)-**4** and (B) (3*R*, 4*R*)-**5**. The omit Fo – Fc density map for each inhibitor is shown at the 3.0 σ contour level. Major hydrogen bonds are depicted with dashed lines.

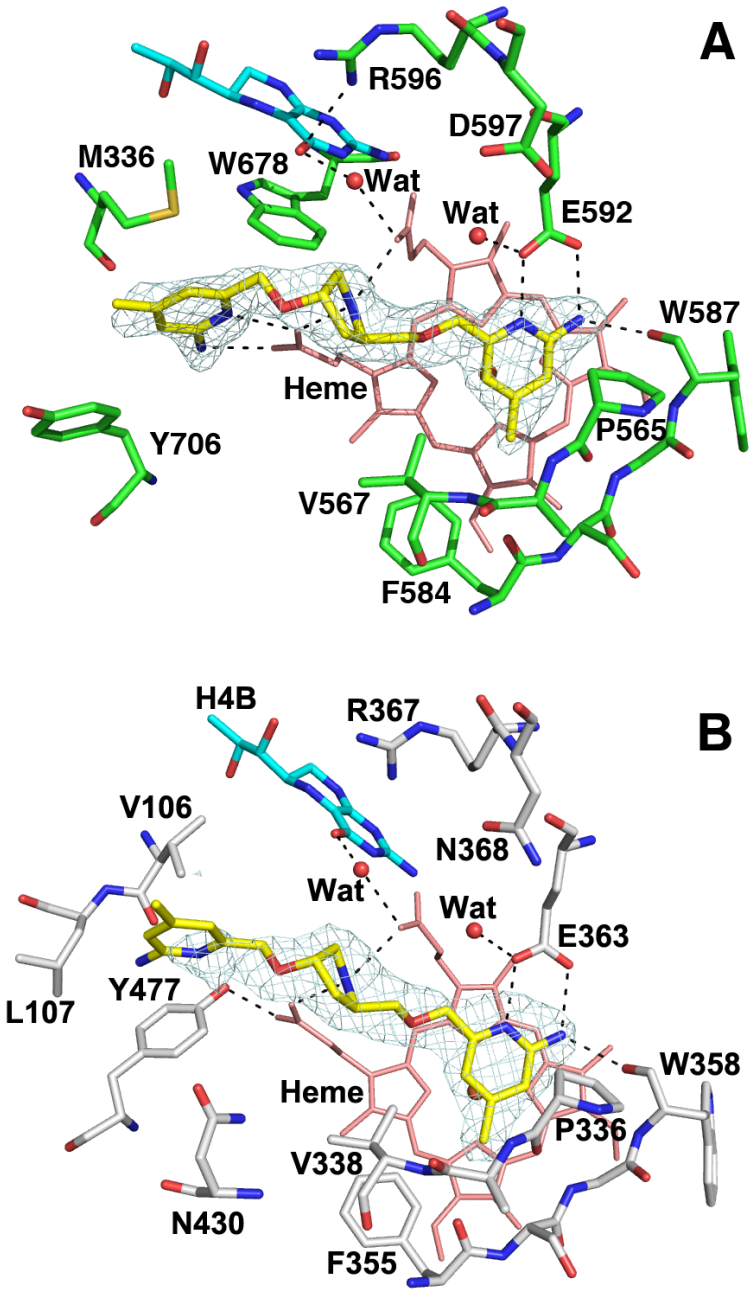


Fig. S4 Compound (2*R*, 4*S*)-**6** bound to the active site of (A) nNOS (4C39, [3]) and (B) eNOS (4C3A, [3]). The omit Fo – Fc density map for each inhibitor is shown at the 3.0 σ contour level. Major hydrogen bonds are depicted with dashed lines.

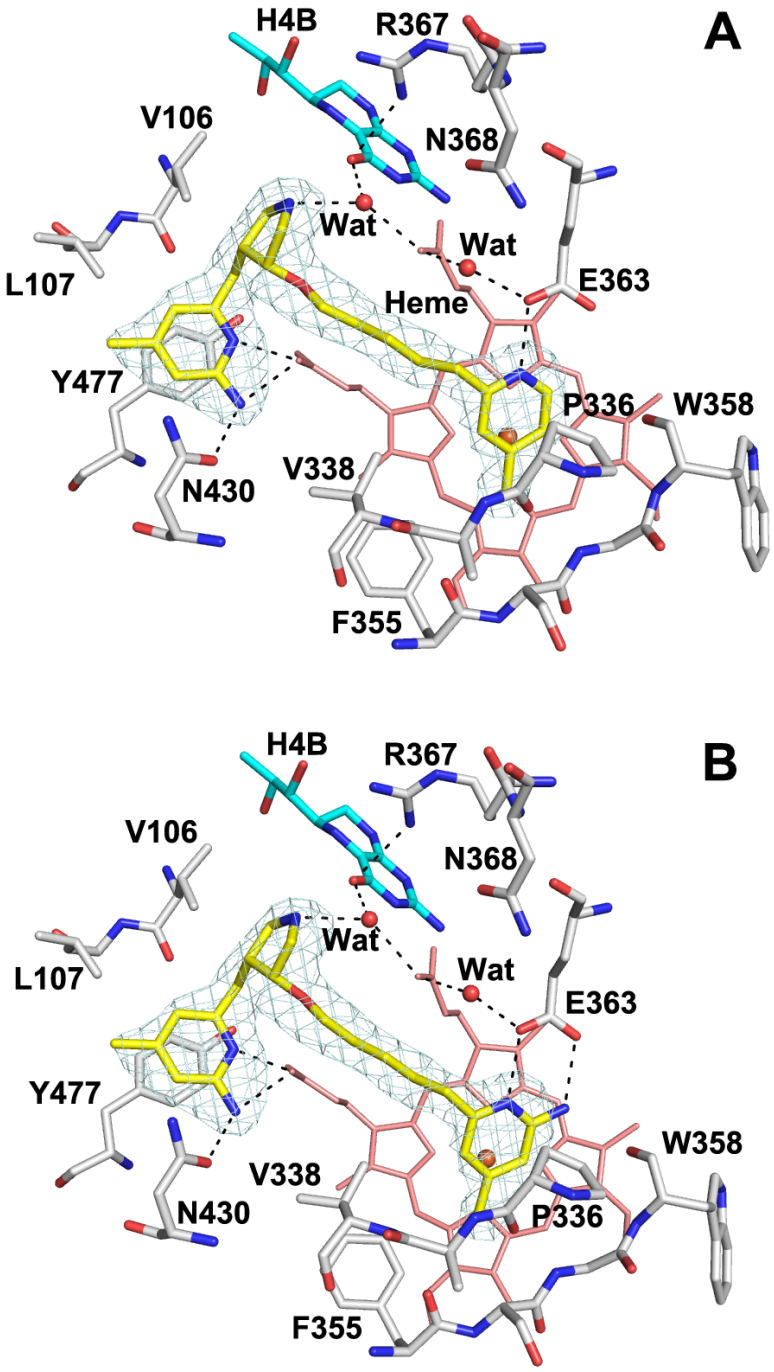


Fig. S5 The eNOS L111A active site bound with (A) (3*R*, 4*R)*-**3** and (B) (3*R*, 4*R)*-**5**. The omit Fo – Fc density map for each inhibitor is shown at the 3.0 σ contour level. Major hydrogen bonds are depicted with dashed lines. The binding modes of (3*R*, 4*R)*-**3** and (3*R*, 4*R)*-**5** are the same as that observed in wild-type eNOS (Fig. 3B and Fig. S2B, respectively).

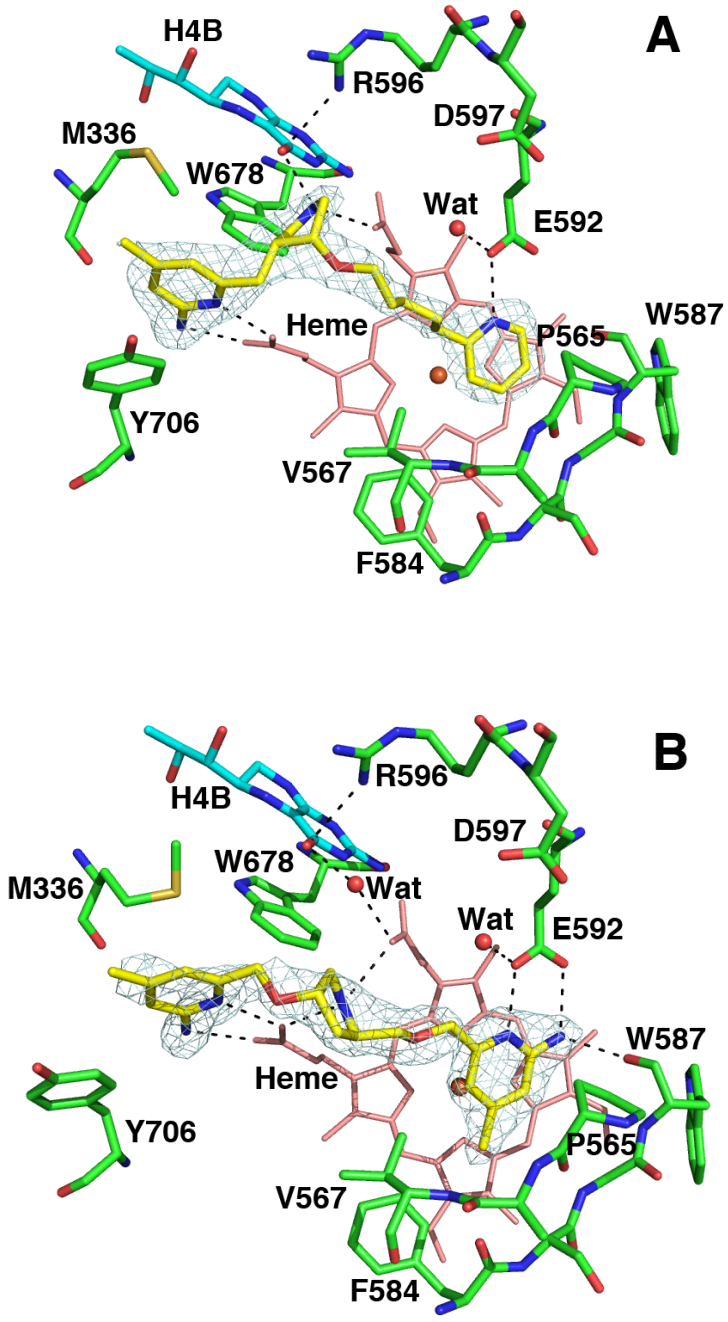


Fig. S6 The nNOS H341L active site bound with (A) (3*R*, 4*R)*-**2** and (B) (2*R*, 4*S)*-**6**. The omit Fo – Fc density map for each inhibitor is shown at the 3.0 σ contour level. Major hydrogen bonds are depicted with dashed lines. The binding modes of **2** and **6** are the same as that seen in wild-type nNOS (Fig. 2A and Fig. S3A, respectively).

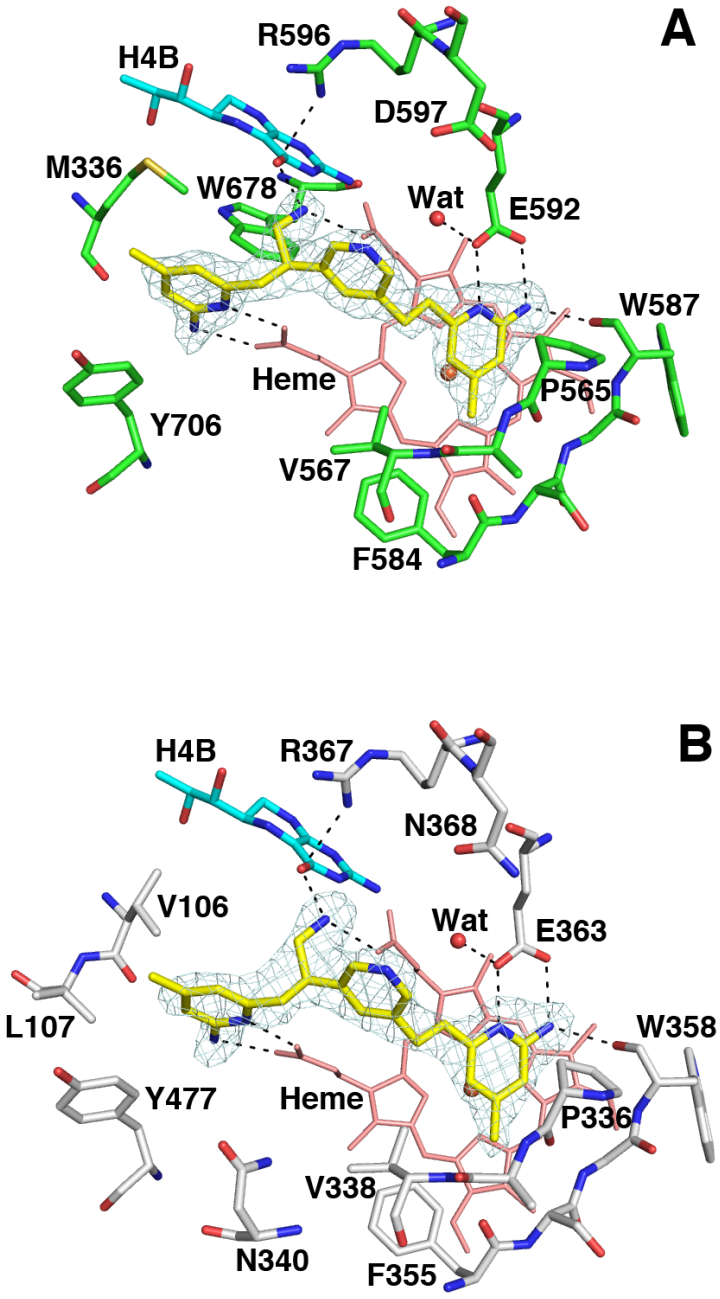


Fig. S7 Compound **7** bound to the active site of (A) nNOS (4CTW [4]) and (B) eNOS (4CU0 [4]). The omit Fo – Fc density map for each inhibitor is shown at the 3.0 σ contour level. Major hydrogen bonds are depicted with dashed lines.

References

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(2) Huang, H., Ji, H., Li, H., Jing, Q., Labby, K.J., Martasek, P., Roman, L.J., Poulos, T.L., and Silverman, R.B. (2012) Selective monocationic inhibitors of neuronal nitric oxide synthase. Binding mode insights from molecular dynamics simulations. *J Am Chem Soc 134,* 11559-11572.

(3) Jing, Q., Li, H., Roman, L.J., Martasek, P., Poulos, T.L., and Silverman, R.B. (2014) An Accessible Chiral Linker to Enhance Potency and Selectivity of Neuronal Nitric Oxide Synthase Inhibitors. *ACS Med Chem Lett 5,* 56-60.

(4) Kang, S., Tang, W., Li, H., Chreifi, G., Martasek, P., Roman, L.J., Poulos, T.L., and Silverman, R.B. (2014) Nitric oxide synthase inhibitors that interact with both heme propionate and tetrahydrobiopterin show high isoform selectivity. *J Med Chem 57,* 4282-4296.